RU e-st System Specification

Note: The sample in this handout contains only a few pages of the specification.

Disclaimer: This is a fictitious system and does not claim to be a model for a good Software Requirement Specification. It is used for exercise purposes only.

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1. INTRODUCTION

RU Financial Services is currently beginning an e-commerce project to develop software for a Web-based online stock and securities trading system. The new system has been tentatively named the **RU e-st** system.

2. Purpose

The purpose of this SRS is to serve as a statement of understanding between the Online Trading Business and Engineering Divisions of RU Financial Services. Any desired departures from the specifications described herein must be negotiated by the affected parties.
The audience for this SRS includes the engineers who are involved in the development of the system, the marketers who are involved in its advertisement and sale, and the testers who validate the system.

2.1 Scope

The overall objective of the **RU e-st** Stock Trading System is to provide our trading customers with a convenient way to manage their stock portfolios. Managing a stock portfolio includes such capabilities as buying and selling and researching stock in a secure environment.

3. REQUIREMENTS

In this section all of the functions that the **RU e-st** Stock Trading System Software is to perform are specified. These specifications are first given from total system perspective.
The **RU e-st** system shall provide all of the current functionality of the current stock trading system, plus any additions specified in this document.

3.1 Trading Customer Requirements

The **RU e-st** system allows its users to trade securities online, using a Web interface (an existing browser such as Netscape or Internet Explorer). A trading customer can log on to the Internet anywhere such as a hotel room, airport, and so on. Users can perform all the basic operations for securities trading: open accounts, trade stocks and other securities, and transfer assets among RU e-st accounts. Users can also obtain current and historical information about their accounts, such as the number of shares held, the current price, and the account total value.
Customers can execute many types of trade orders: market trading (buy and sell at current market prices), transfers from one mutual fund to another (within one account), and limit trading (buy and sell at a specific price). The **RU e-st** system uses the existing Market Trading System to perform the securities trades. **RU e-st** also allows users to transfer cash in an account to or from financial institutions such as banks or credit card vendors, using the existing Financial Network System. The usual restrictions apply:

- For online market sales, the securities to sell must be in the customer's account.
- For market purchases, cash for the purchase must be in the account by the settlement date.
- Transfers and purchases from an account are allowed only if, at the time the transaction occurs, they have enough cash to fund the transaction in the account.

Trading customers can obtain information about what is happening in the securities markets. A trading customer can obtain price quotes and news headlines by entering the trading symbol (for example, "IBM") for a particular security. The **RU e-st** system obtains current and historical quotes from the existing Quote System service and news items from the existing News System. **RU e-st** also has a feature to broadcast news headlines periodically on the trading customer's screen (without being asked).

3.2 Regulatory Requirements

The system must report yearly tax information to the IRS and state tax boards. Tax forms must be communicated to the IRS and copies mailed to the customer. The information is also available online for customer viewing.

3.3 System Administration Requirements

Updates to the information on the Web pages must be possible without making the system unavailable.

3.4 Development Requirements

The system must be written in one of the company's approved programming languages. Refer to document

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Enterprise Architecture and Development Standards document EA-1234[1] for details.

3.5 Hardware Requirements

The hardware platform must be one supported by the enterprise hardware maintenance contract. Supported platforms are specified in the Enterprise Architecture and Development Standards document EA-1234.

[1] This document is not supplied with the course.

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Use-Case Model Survey

RU e-st System

Introduction

This set of sample solutions is for the RU e-st system for Web-based stock trading. The point of this set of solutions is to show what a solution may look like and to what level of detail you describe the use cases. The stock trading system is based on a real system. However, details of stock trading have been simplified at many points in order to make a case study suitable for discussion in a relatively short course. The point here is not the technical content but what the descriptions may look like.

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1. Introduction

This is an example of what a solution to the use-case exercises in the MRMUC course might look like. It contains more information than a MRMUC class might create in the limited class time available. It is an example of a use-case model for a regular project. Keep in mind that this example shows the use-case model at the beginning of a project.

2. Survey Description

The RU e-st system allows its users to trade securities online, using a Web interface (an existing browser such as Netscape or Internet Explorer). A trading customer can log on to the Internet anywhere – hotel room, airport, and so on. Users can perform all the basic operations for securities trading: open accounts, trade stocks and other securities, and transfer assets among RU e-st accounts. Users can also obtain current and historical information about their accounts, such as the number of shares held, the current unit price, and the current total value. Trading customers can obtain information about what is happening in the securities markets, such as current price quotes and news items. In addition, the system must report yearly tax information to the IRS and state tax boards.

3. Actors

3.1 Diagram

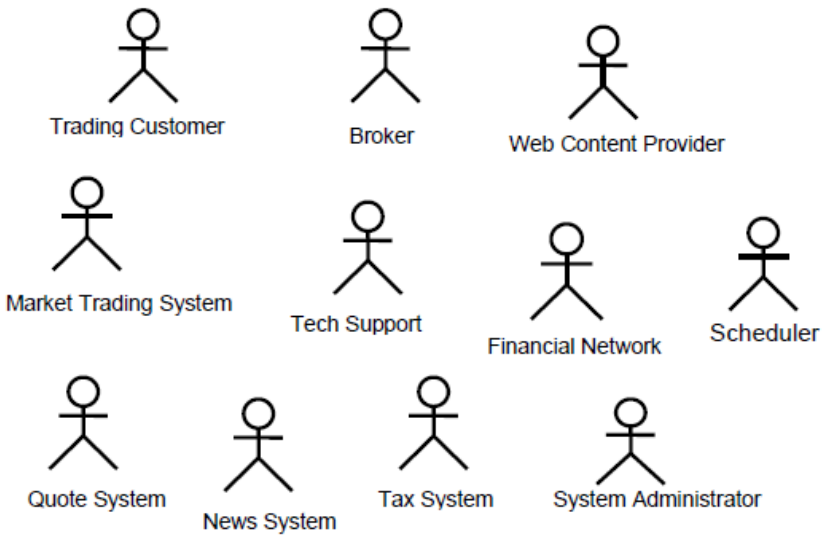


Figure 1. Actor Diagram

3.2 Trading Customer

A person with a valid customer id and password. The Trading Customer is the primary end-user who trades securities (stocks and mutual funds) through the RU e-st system.

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3.3 Broker

The Broker has access to accounts assigned to the broker.

3.4 System Administrator

The System Administrator has responsibility for the entire RU e-st system. The System Administrator authorizes Trading Customers and their trading accounts, authorizes non-standard trades, handles fraudulent use of the system, and monitors the system's overall performance and health.

3.5 Customer Service

Customer Service has responsibility for the operation of the RU e-st system. Customer Service answers questions from Trading customers based on RU e-st account information and accesses and resolves problems with individual accounts.

3.6 Web Content Provider

The Web Content Provider has responsibility for posting broadcast and other messages on the Web site for the RU e-st system.

3.7 Market Trading System

The system performs the market trades requested by RU e-st and communicates trade results back.

3.8 Financial Network

A Financial Network provides services to the RU e-st system and is responsible for verifying accounts at banks, authorizing transactions, and recording completed transactions. The Financial Network makes it possible for Trading Customers to transfer funds between RU e-st accounts and accounts external to RU e-st, such as bank accounts or credit card accounts.

3.9 Scheduler

A human or system that initiates a use case based on a calendar event.

3.10 Tax System

This system accepts electronic reports of tax information.

3.11 Quote System

This system provides current and historical market price quotes of securities such as stocks and mutual funds.

3.12 News System

This system provides news headlines about a particular security (stock or mutual fund).

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4 Use Cases

4.1 Primary Cases

4.1.1 Diagram of Use Cases Around the Trading Customer

See Figure 2.

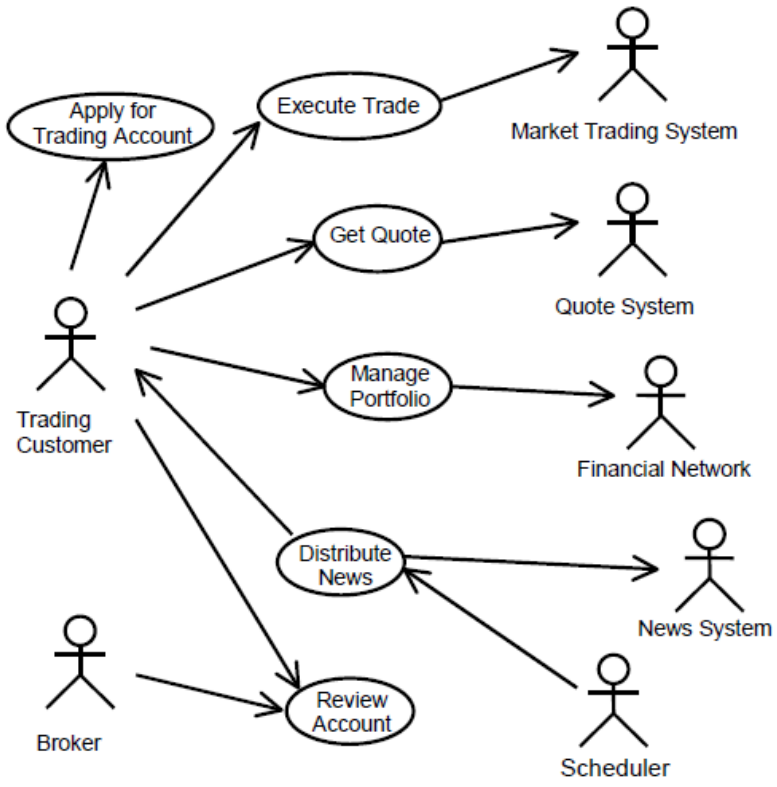


Figure 2. Diagram of Use Cases around the Trading Customer

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4.1.2 Apply for Trading Account

A person can apply for a new trading account. If the person does not already have any trading accounts, the person is authorized to become a new Trading Customer. A Trading Customer with existing trading accounts can set up new trading accounts of the same or different account type.

4.1.3 Execute Trade

Trading Customers buy, sell, or transfer securities in their accounts. Immediately after a trade is completed, Trading Customers receive a market trading confirmation containing a transaction id and the trade results.

4.1.4 Manage Portfolio

Trading Customers can manage their portfolios. Allowable transactions include transferring cash or other assets between accounts, reviewing accounts, consolidating or splitting accounts, and transferring cash between their RU e-st accounts and accounts at banks or credit card companies.

4.1.5 Get Quote

Trading Customers get current and historical information about the trading price of securities.

4.1.6 Review Account

Trading Customers get current and historical information about their accounts. Brokers can also use this use case to review accounts that belong to Trading Customers whose accounts they help manage.

4.1.7 Distribute News

The RU e-st system periodically broadcasts headline news and current stock information and trading tips.

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4.2 Secondary cases

4.2.1 Diagram of Support and Maintenance Use Cases

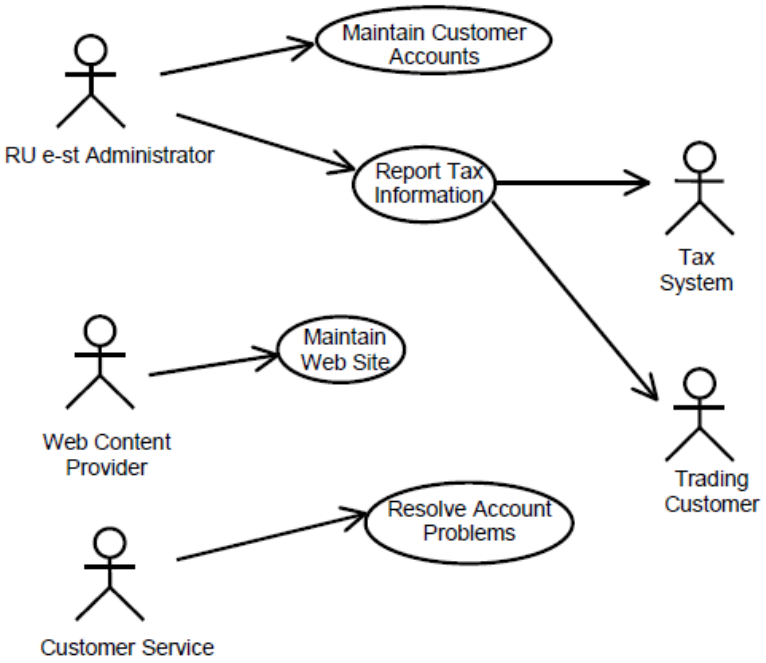


Figure 3. Diagram of Support and Maintenance Use Cases

4.2.2 Maintain Customer Accounts

The RU e-st Administrator uses the Maintain Customer Accounts use case to authorize Trading Customers and accounts, authorize non-standard trades, and oversee the administration of the RU e-st system at a given financial institution.

4.2.3 Report Tax Information

The RU e-st Administrator uses the Report Tax Information use case to report tax information to the Tax System. Both state and U.S. taxes can be reported.

4.2.4 Maintain Web Site

The Web Content Provider uses the Maintain Web Site use case to change the Web content: modify messages for broadcast, modify screen content, and modify links among Web pages.

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4.2.5 Resolve Account Problems

Customer Service uses the Resolve Account Problems use case to examine and change customer accounts and transactions. Customer Service representatives communicate with Trading Customers by e-mail and telephone (outside of the RU e-st system) about problems with accounts or transactions and use RU e-st to resolve those problems.

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Get Quote Use Case

1. Brief Description

Trading Customers get current and historical information about the trading price of securities.

2. Flow of Events

2.1 Basic Flow

1. Customer logs on.

2. Customer selects "Get Quote" function.

3. Customer selects stock trading symbol.

4. Get desired quote from Quote System.

5. Display quote.

6. Customer logs off.

2.2 Alternative Flows

A1 Customer Gets Other Quotes
A2 Unidentified Trading Customer
A3 Quote System Unavailable
A4 Quit
A5 Unknown Trading Symbol

3. Scenarios

3.1 Success Scenarios

Customer gets a quote: Basic Flow
Customer gets additional quotes: Basic Flow, Customer Gets Other Quotes

3.2 Failure Scenarios

Invalid login: Basic Flow, Unidentified Trading Customer
Quote system host link down: Basic Flow, Quote System Unavailable
Trading customer enters an unknown trading symbol: Basic Flow, Unknown Trading Symbol
Trading customer quits before completion: Basic Flow, Quit

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Execute Trade Use Case

1. Brief Description

Trading Customers buy, sell, or transfer securities in their accounts. The possible trade order types are: Limit Buy Order, Limit Sell Order, Market Buy Order, Market Sell Order, Transfer Order. Immediately after a trade is completed, Trading Customers receive a market trading confirmation containing a transaction id and the results of the trade.

2. Flow of Events

2.1 Basic Flow

1. Customer logs on.

2. Customer selects "Trade" function.

3. Customer selects account.

4. Customer performs trade.

    4.1 Select trade order type (market, limit, transfer) and enter trading info.

    4.2 Initiate trade with Market Trading System.

    4.3 Get trade results from Market Trading System.

5. Customer logs off.

2.2 Alternative Flows

A1 Unidentified Trading Customer
A2 Unauthorized Trader
A3 Quit
A4 Trade Is Rejected
A5 Market Trading System Unavailable
A6 Insufficient Funds
A7 No Trading Account
A8 Account Not Operational
A9 Customer Executes More Trades

3. Scenarios

3.1 Success Scenarios

Customer executes a trade: Basic Flow.
Customer Executes More Trades: Basic Flow, Customer Executes More Trades.

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3.2 Failure Scenarios

Invalid login: Basic Flow, Unidentified Trading Customer.
Trader is not authorized to use account: Basic Flow, Unauthorized Trader.
Customer quits before executing trade: Basic Flow, Quit.
Trade rejected: Basic Flow, Trade Is Rejected.
Host link down to Market Trading System: Basic Flow, Market Trading System Unavailable.
Insufficient funds in trading account: Basic Flow, Verify That Cash Is Available.
Customer doesn't have a trading account: Basic Flow, No Trading Account.
Trading account is inactive: Basic Flow, Account Not Operational.

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